

UNIVERSITY OF TORONTO

ECE496 IMPLEMENTATION PLAN

TEAM 2025162, SECTION 2

**Vision-Guided
Automated Foosball Opponent**

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1.0 Summary of Project Status and Changes

The following section provides an overview of the current project status and changes made since the Proposal.

1.1 Implementation Progress

The software side of the team is transitioning from a small prototype foosball table to a full-size one. A 100 FPS global shutter color USB camera (ArduCam OV9782 [1]) was acquired, satisfying all requirements for the input device in Section 4.1. A stand for the camera module was made to ensure the playing field is visible in the frame, 3D printed for the prototype, but built with wood for the full-size. The software now has a GUI to view detections of the game state and signals sent to the motors, and can detect the ball's position using color filtering, extrapolating its trajectory to rod interception points. This runs on a newly acquired Raspberry Pi 5 [2] with the camera connected. On the mechatronics side, the chosen motor (StepperOnline iSV57T-180S [3]) was tested for both rotational and translational functionalities. To power them, we used a MEAN WELL LRS-350-36 [4] for which we self wired and 3D printed a custom enclosure. The translational motor was connected to a belt-driven linear actuator (HPVB45 [5]), whose carriage system had the rotational motor attached on it. The overall motor setup required custom 3D printed components which our team designed; this allowed for proper load and inertia testing on both motors, aligning with the calculations in Appendices D and E. A CAD model of our wooden stand, power supply, and motor setup can be seen in Appendix F.

1.2 Challenges Encountered

The software side assumed that real-time positions of the motorized rods would be available through an interface with the motors, making requirements for foosball rod position detection more lax (only the non-motorized side of the table would need this technique, and accuracy can be lower, as the purpose would be for strategy in offensive play with no effect on the motorized side's ability to deflect the ball). However, this is no longer the case with the chosen motors. Thus, the importance of accurate color filtering (or alternative techniques) has been increased, for all directions and intensities of light. Another method to gather rotational positions of the rods (not visible to the camera) is needed. The mechatronics side found challenge in the coupler's connection to the motor. Because the iSV57T180's output shaft is milled in a D-shape, it did not fit snugly into the circular bore of the coupler, resulting in a loose connection. This issue required a lot of careful alignment and tightening to resolve. Overall, extensive calculations and research gave the team confidence on most hardware components, allowing the power supply and motor setups to be built smoothly.

1.3 Plans Ahead

After verifying existing functionalities of the software on the full-size foosball table following the installation of the camera and microcontroller, the software side plans to add lateral rod position detection functionality, improve stability in lighting of the frame, and explore solutions for detecting rotational rod position before installing motor control logic. The mechatronics side will be connecting and coding a Teknic ClearCore [6] motor controller to control all motors simultaneously. Furthermore, we plan to purchase an additional table to place the hardware components on the side of the foosball table for full assembly and proper testing purposes.

2.0 Possible Solutions and Design Alternatives

This section details alternative design solutions and elaborates on why they were rejected.

2.1 List of Considered Alternatives

Before finalizing our chosen design, three system-level alternatives were explored to evaluate different approaches to vision processing and ball tracking for the automated foosball opponent seen in Table 1.

Table 1. Considered system-level alternative designs and brief reasons for why they were rejected.

ID	Description	Reasoning for Rejection
A	Full-size desktop with a GPU paired with existing color camera using a deep-learning architecture to identify ball and/or player positions	Excessively costly and power-intensive for the intended application. FPS is very low [7] and rotational angle difficult to calculate.
B	Existing microcontroller paired with monochrome camera that uses purely edge-detection mechanisms to identify ball position and player states	Lacks robustness and computational capability. Also easily foiled by other debris that can make its way onto the table, since the camera cannot distinguish between colors.
C	Avoiding vision usage entirely and instead relying on lasers on the table or GPS sensor in ball to detect its position and accelerometers on all rods to determine position	Excessive hardware complexity, calibration challenges, and integration issues.

Alternative A proposes a full-size desktop-based system utilizing a GPU paired with a color camera and deep-learning models for visual recognition of the ball and players, offering potentially high accuracy at the expense of cost, power, and complexity. Alternative B considered a lightweight microcontroller solution using a monochrome camera and edge-detection algorithms to identify ball position, trading computational simplicity for limited robustness and adaptability to changing lighting conditions. Alternative C eliminated vision altogether by relying on external sensors—such as lasers, GPS-like tracking within the ball, and accelerometers on the rods—to determine position data, introducing considerable hardware complexity and calibration challenges. Each design explored a different trade-off between computational power, sensing accuracy, and integration complexity.

2.2 Elaboration and Comparison of Alternative A

Alternative A: Full-size desktop with a GPU paired with an existing color camera using a deep-learning architecture to identify ball and/or player positions.

This alternative explores the use of a desktop-class computer equipped with a discrete GPU to perform deep-learning-based vision inference. In theory, such a system could run convolutional neural networks (CNNs) or transformer-based detectors to recognize and track the foosball, players, and rods in real time.

The large compute headroom of a GPU would allow for high-resolution input and potentially robust object detection even under challenging lighting or occlusion conditions.

However, several key drawbacks make this option unsuitable for the intended design objectives. A full desktop introduces significant cost, size, and power consumption far exceeding the project’s constraints [8]. The setup would require multiple pieces of large hardware, and make the already tight setup space even less achievable. Deep-learning pipelines also demand substantial training data collection and model tuning to handle variations in table color, lighting, and camera placement—an impractical effort for a prototype-scale project [9]. We would have to adjust lighting in multiple locations, but moving the table for every scenario and occasion is impractical. Moreover, GPU-based inference produces unnecessary overhead for tracking a single small object, while increasing latency due to data transfer and preprocessing steps. These limitations reduce maintainability, portability, and reliability, making the system harder to demonstrate and deploy.

Comparison with Implemented Approach: In contrast, the implemented approach utilizes a Raspberry Pi 5 running optimized classical computer-vision algorithms such as color segmentation, thresholding, and ArUco marker detection. This design achieves real-time tracking using lightweight computation that fits within the Pi’s limited CPU resources [9]. The choice of traditional vision techniques provides fast inference without the need for model training, while the use of ArUco markers offers reliable and repeatable localization of player rods and ball position under varying lighting conditions.

Compared to Alternative A, the implemented system is:

- Far more cost-effective, requiring only a single-board computer and a low-cost camera rather than a GPU-equipped PC.
- More compact and portable, allowing the vision module to mount directly on or near the foosball table without external peripherals.
- Simpler to develop and maintain, as it relies on open-source computer-vision libraries (OpenCV) [10] and deterministic algorithms instead of complex neural-network pipelines.
- Lower power and quieter, with negligible heat generation and no need for active cooling.
- Sufficiently accurate for tracking purposes: while it may not match the theoretical precision of a trained CNN under perfect conditions, it achieves robust, consistent detection that meets gameplay and control requirements [9].
- Can run at a much higher frame rate (FPS) compared to deep-learning–based approaches, even powered under a GPU [7].

Overall, while Alternative A represents the most powerful computational option, its resource demands and engineering complexity make it over-engineered for the problem. The chosen Raspberry Pi–based vision system delivers an optimal balance of performance, cost, and simplicity, fulfilling all design requirements within practical constraints. It demonstrates that lightweight, well-tuned image-processing methods can achieve reliable real-time tracking without resorting to deep-learning hardware or training overhead.

3.0 Technical Design and Implementation

This section will discuss the entire design plan, at a high-level and further broken down into sub-modules.

3.1 System-Level Overview

The overall goal of the design is to transform a video feed of the foosball playing field into the movement of two rods that intercepts an incoming ball. The process flow begins with a camera positioned overhead the table to record a bird's-eye-view of the table. The camera feed is then processed through a Raspberry Pi 5 Single Board Computer (SBC). The SBC will initially gauge the boundaries of the table and the axes corresponding to each of the rods using the existing method of ArUco markers ("synthetic square marker composed by a wide black border and an inner binary matrix which determines its identifier (id) [11]") for perspective transformation. Color filtering techniques in OpenCV commonly found in internet guides such as [12] are used to acquire ball and foosmen positions, through which the trajectory of the ball is calculated and points of interception are found on the rods, which then become the desired linear position for each of the rods. A stretch goal will involve a novel algorithm to calculate the angle of attack to line the ball through empty spaces of the opponent's foosmen.

Once the rotational angle and the linear position for each of the two rods is known, the motors will be commanded to spin to meet these targets. For each rod, one motor controls a linear guide rail, using the motor's rotation to move a platform across a belt. This platform holds the other motor, which rotates the rod through a direct attachment. The motors used are contained within integrated servos, housing a driver controlled through voltage pulses. The Teknic ClearCore will be used to manage the direct signal control, allowing use of its API to program the servos [6]. This was chosen for its speed and reliability as opposed to a microcontroller, which would also be more prone to errors in programming the timing. The SBC, motors, and controller are all powered through connection to an AC outlet. Distance is not an issue, however, as an extension cord with 6 outlets will be used to connect each power supply individually. Except for the SBC, the components use switching power supplies configured for individual DC voltage requirements.

3.2 System Block Diagram

Figure 1 visually illustrates the system's modules and connections, in addition to the separation of tasks. Note that the diagram and the following explanations assume a system with 1 rod actuated. If funding can be acquired for a 2nd rod, the control unit will essentially be duplicated.

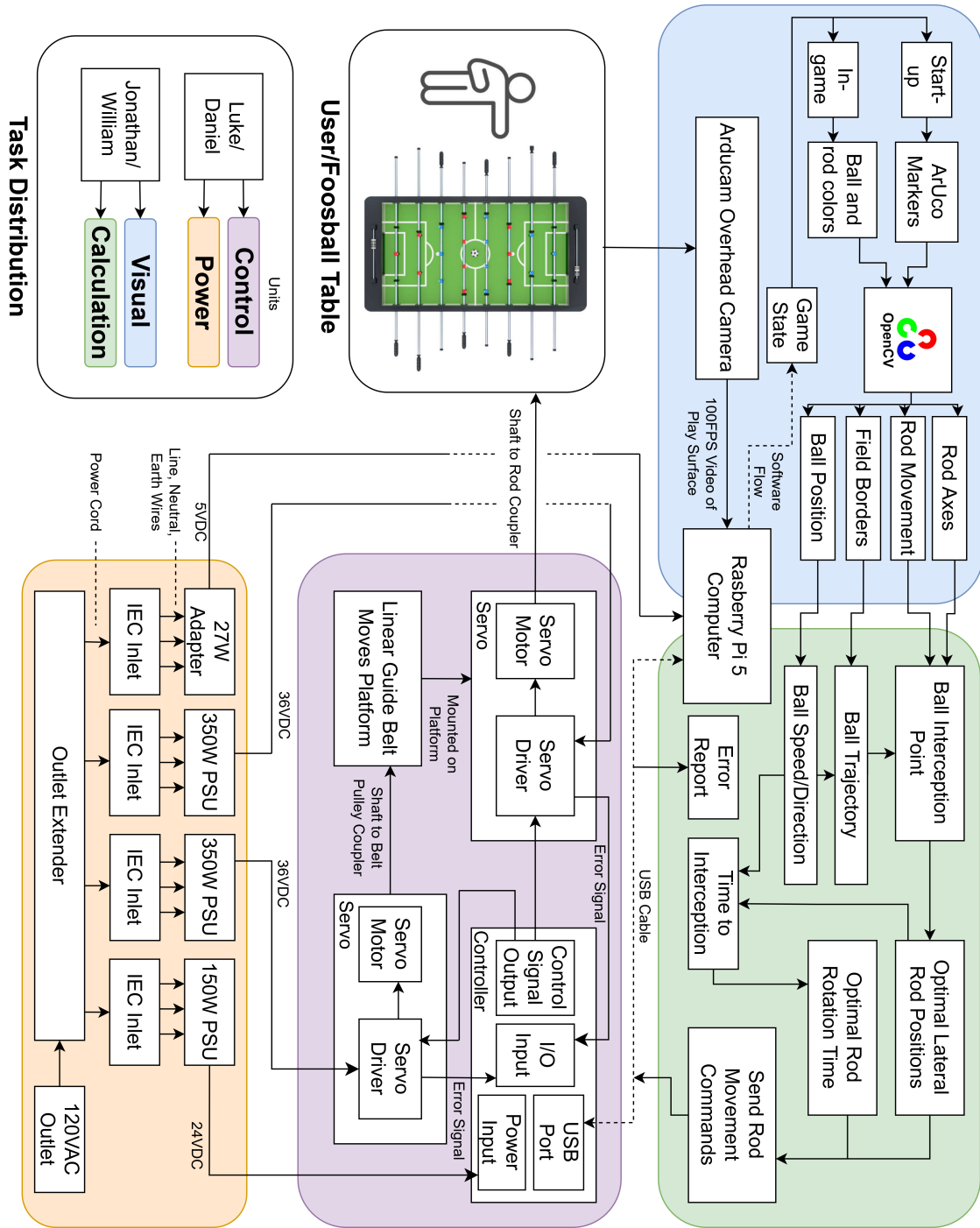


Figure 1. System block diagram.

3.3 Module Descriptions

This subsection explains the role of each connection and unit from our system block diagram.

3.3.1 Input/Output

The user interacts with one side of the foosball table, which has no modifications to it. The other side of the table is connected to our system and provides the rod(s) with movement response for the user to interact with.

3.3.2 Visual Unit

The Arducam overhead camera, after taking in a frame of the whole playing surface, sends in the frame to a Raspberry Pi 5 computer, which, on startup, detects ArUco markers on the table through the OpenCV library to gain information about the axes of the rods and border of the playing field. In game, the detection is instead for the colors of the ball and of the markers on the rods, also through OpenCV, to calculate information on the ball's position and rods' translational movements. This information is handed to the Calculation Unit.

3.3.3 Calculation Unit

Changes in the ball's position are tracked to acquire its speed and direction, which is then extrapolated, including bounces off the playing field's borders, then combined with rod axes locations to find interception points and thus the optimal lateral positions for each rod. Any lateral movements required to reach the positions are requested. The time taken for the ball to reach interception is also calculated using the current speed on the length of the trajectory. If that time is within the optimal rod rotation time, it requests a rotation (a "kick"). Once a rod rotation or lateral movement is requested, it is first translated to a servo movement command using the ClearCore API which is then sent over a USB serial to ClearCore controller.

3.3.4 Control Unit

When the ClearCore controller receives a command to move a servo, it will translate that request to a voltage signal output to the servo's driver. The driver also sends a signal back to the controller if a severe failure occurs, which when received by the SBC can terminate further commands. One servo is responsible for moving the belt on the linear guide rail through a pulley system, attached to the servo through a shaft coupler. The other servo is mounted on the belt's carriage and directly connected to a rod through an additional coupler. This servo will rotate the rod and pull the rod with it when it moves linearly on the guide rail.

3.3.5 Power Unit

This section encompasses the components that deliver power to the rest of the system. There are four power supplies needed, two for the servos, one for the controller, and one for the SBC. It is indeed unnecessary to have individual power supplies for each unit, however this was found to be much cheaper as opposed to consolidating into one power supply. Aside from the 27W Adapter for the SBC, which resembles a laptop charging device, the rest are switching supplies with two delivering 36VDC for the servos and one 24VDC for the controller. These supplies have exposed terminals, so a 3D printed case has been used to allow for power cords to connect to inlets on the case, which are wired to the terminals inside.

4.0 Test and Verification

This section elaborates on methods to test and verify the design, for the requirements specified in Appendix A for the two major components: vision/strategy and mechatronics. Completed verifications are marked with (C).

4.1 Vision and Strategy

Table 2 details the testing methodology for the vision and strategy components of the design.

Table 2. SyRS of the vision/strategy component with testing methodologies.

ID	Specification	Verification Strategy
1 (C)	Input device framerate > 60 FPS.	REVIEW of DESIGN: Reference the datasheet of visual input unit for FPS. TEST: Verify input framerate through an application that repeats unprocessed frame input to the GUI.
2 (C)	Input device must detect in full color.	TEST: Using a webcam application that only repeats frame input to the GUI, confirm the camera is not monochrome.
3	Calculation of required rod movement for all motorized rods in under $\frac{1}{input\,framerate}$ seconds.	TEST: Add an FPS counter to the run script that displays with the debug mode (see requirement 5), where the "FPS" is calculated by the times each processed frame is displayed.
4 (C)	Input device must take all areas of frame simultaneously.	REVIEW of DESIGN: Reference the datasheet of visual input unit. Confirm it has a global shutter.
5 (C)	Software must have a visual "debug" mode which shows detections, activations, and ball motion state.	TEST: Confirm run with a flag on the run script which enables a live view of the camera feed in the GUI with overlays of required information, set by default to off.
6	Device must be able to detect whole ball and player positions in each frame	TEST: In debug mode (see requirement 5), there must be one outline of the ball, and one outline per synchronization point for the lateral positions of each rod and foosmen.
7 (C)	Input device must not be visually obstructed on the way to the ball or the players	TEST: Once visual input unit and controller is installed on the foosball table, confirm there is nothing blocking view of the playing surface through an application that only repeats the frame input to the GUI.
8 (C)	Input device must connect using a universal interface (e.g. USB).	REVIEW of DESIGN: Reference the datasheet of visual input unit and confirm it utilizes a universal interface, and demonstrate above requirements while using said interface.
19 (C)	(NEW) Height of support structure must be enough to allow for selected camera to contain entire field	TEST: Confirm the entire playing surface, including borders, is visible through an application that only repeats the frame input to the GUI.
20	(NEW) Selected computer must be able to run the software in real-time	TEST: Confirm the software can perform all functions on the chosen SBC.

4.2 Mechatronics

Table 3 details the testing methodology for the control and power components of the design.

Table 3. SyRS of the mechatronics component with testing methodologies.

ID	Specification	Verification Strategy
9,22 (C)	Input Power: Minimum 24VDC/7.5A for motors, 24VDC/5A for controller/	TEST: Measure the output of the acquired power supplies before connecting them to the motors/controller.
11,18	Construction: System does not make permanent modifications to the foosball table and does not obstruct user side.	REVIEW OF DESIGN: Confirm whether the completed system be easily removed from the table such that the original use and structure of the table is intact, and when it is present it does not obstruct unmodified side.
12 (C)	Rotational Motor Specifications: The motor must rotate at least 120 degrees, and output peak torque and speed of 1.158Nm and 1000RPM.	TEST/REVIEW OF DESIGN: The selected motors meet these specifications and have been tested to rotate the rod.
13 (C)	Translational Motor Specifications: The motor must support up to 2kg in translational movement, rotate 360 degrees, and output a peak torque and speed of at least 1.2Nm and 500RPM.	TEST/REVIEW OF DESIGN: The selected motors meet these specifications and have been tested with the target load connected to the guide rail.
14 (C)	Linear Guide Rail: The maximum travel length is 400mm (measured internally by the team).	TEST: The selected guide rail has been tested with the selected motors to achieve this travel length.
15,17	Defensive Ability: Can defend for at least 1 minute and react in 0.2s	TEST: Time the completed system to stop an untrained player from scoring a goal for at least one minute, also measuring rod movement time using the camera.
21 (C)	Controller must have 4 outputs, each with 5V step and direction signals at 300KHz per servo requirements.	REVIEW OF DESIGN: Select component exceeds these specifications.

5.0 References

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6.0 Appendices

Additional relevant research information and calculations are listed in their respective appendix sections below.

6.1 Appendix A: System Requirements Specification (SyRS)

Table 4 shows the systems requirements for the vision and strategy aspects of our design while Table 5 shows the systems requirements for the mechatronic aspects of our design.

Table 4. Requirements, constraints, and objectives of the vision/strategy portion.

ID	Specification	Description
1	Input device framerate > 60 FPS.	Objective: Higher is better.
2	Input device must detect in full color.	Requirement: For more effective detection of objects by color.
3	Calculation of required rod movement including lateral position and kick trigger for all motorized rods in $< \frac{1}{input\,framerate}$ seconds.	Requirement: Each frame should have its own calculation to update the rod position with the latest information.
4	Input device must take all areas of frame simultaneously.	Requirement: Each part of the frame should be equally accurate to the time the information was processed.
5	Software must have a visual "debug" mode which shows detections, activations, and ball motion state.	Requirement: The automated system's vision processing status should have a visible mode for finetuning purposes that can be switched off for optimization when running in normal use.
6	Device must be able to detect whole ball and player positions in each frame	Requirement: Position of ball is required for defensive actions, and player positions for strategic offensive actions.
7	Input device must not be visually obstructed on the way to the ball or the players	Requirement: Cannot see game state if game state is visually obstructed.
8	Input device must connect using a universal interface (e.g. USB).	Constraint: Control hardware will change between simulation and action, and a common interface should be used for the software to use the same structure of input.
19	(NEW) Height of support structure must be enough to allow for selected camera to contain entire field.	Requirement: Cannot determine game state without all portions of the playing field, including positions of the borders.
20	(NEW) Selected computer must be able to run the software in real-time	Requirement: Cannot use a computer that is unable to run the software.

Table 5. Requirements, constraints, and objectives of the mechatronics portion.

ID	Specification	Description
9	Motor Input Power: Minimum 24VDC and 7.5A	Requirement: Power supply must meet this specification. See Appendix G for further details.
10	Components Price: Does not exceed an investment of \$500 from each group member, we must assume there will not be any re-reimbursements.	Constraint: Total cost cannot exceed this amount, therefore we will not acquire top-of-the-line components.
11	Construction: System does not make permanent modifications to the foosball table, rather acts as an accessory that can be removed.	Objective: The original foosball table should remain intact and usable.
12	Rotational Motor Specifications: The motor must rotate at least 120 degrees, and output peak torque and speed of 1.158Nm and 1000RPM.	Requirement: This is for realistic gameplay. See Appendix D for detailed calculations.
13	Translational Motor Specifications: The motor must support up to 2kg, rotate 360 degrees, and output peak torque and speed of at least 1.2Nm and 500RPM.	Requirement: This is for realistic gameplay. See Appendix E for detailed calculations.
14	Linear Guide Rail: The maximum travel length is 400mm (measured internally by the team).	Requirement: The travel distance must be within 400mm.
15	Reaction Time: The reaction time of the motor together should average at 0.2s.	Objective: An average human reaction time is around 0.25s [13], therefore, the estimated motor reaction time needs to be faster to account for the computer processing time for our visual perception setup.
16	Automated Rods: Two or more rods.	Objective: Setup will have two or more automated rods
17	Defensive Ability: One or more minutes.	Objective: The system will be able to defend against a goal for at least one minute against an untrained player.
18	Does Not Obstruct Human Side: No interference.	Requirement: Does not interfere with the opposing side of the foosball table or the abilities of the human to play.
21	(NEW) Controller must have at least 4 independent motor control outputs, each supporting 5V step and direction signals at 300KHz.	Requirement: Controller must meet these specifications to control selected servos [3].
22	(NEW) Controller Input Power: Minimum 24VDC and 5A.	Requirement: Power supply for the controller must meet this specification [6].

6.2 Appendix B: Gantt Chart

Figure 2 on the following page shows the projected timeline of tasks, key deliverables, and major milestones displayed in a Gantt chart.

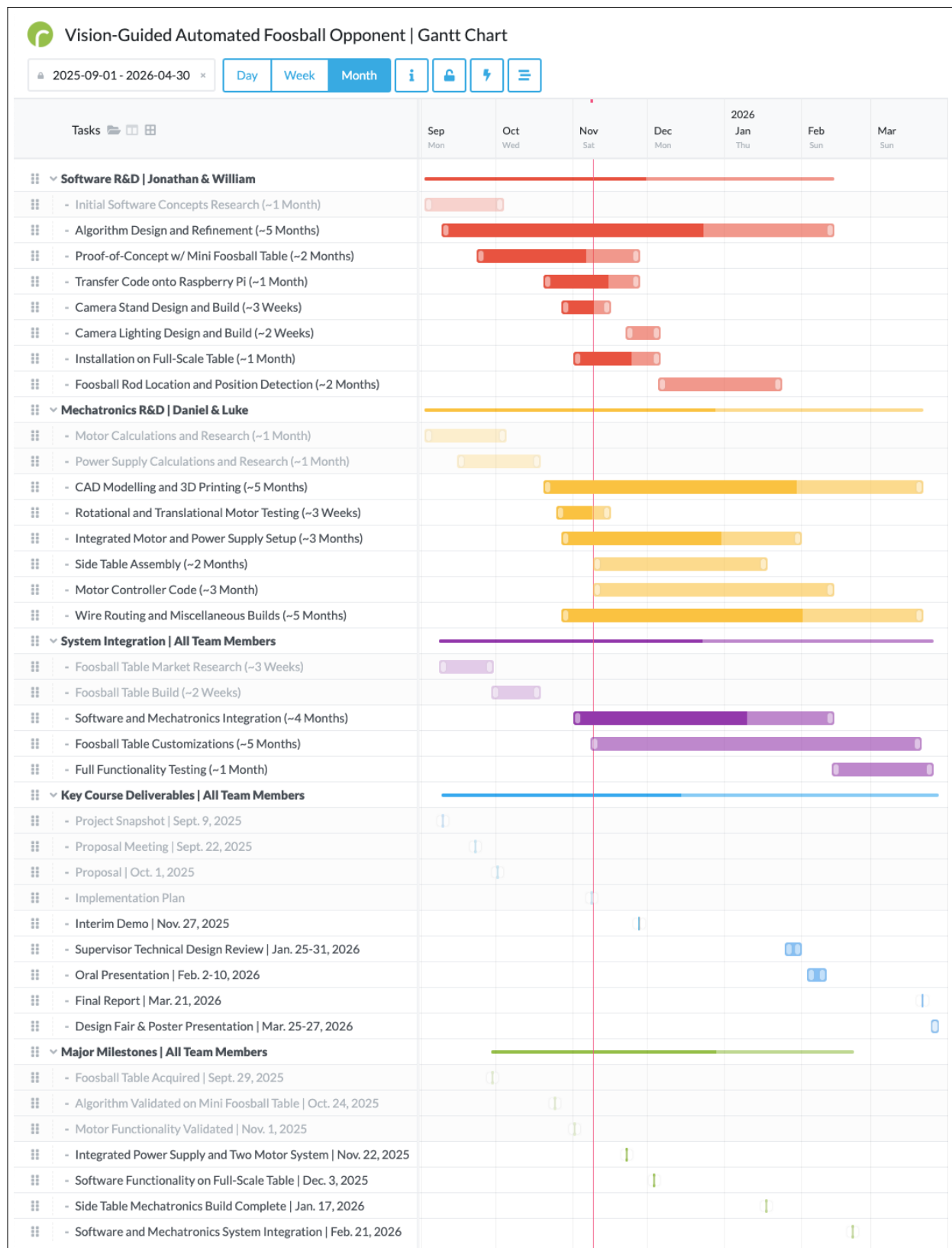


Figure 2. Gantt chart timeline for the 2025-2026 academic year.

6.3 Appendix C: Financial Plan

This appendix details the financial planning for our design and our contingency plans.

6.3.1 Budget Table

Tables 6-8 below details our overall budget plan for our team.

Table 6. Consumables and services budget table.

Consumables / Services	Priority	Cost / Unit	Quantity	Total Cost (\$)	Requires Funding (y/n)
Balls for testing	1	\$7.34	2	\$14.68	y
Miniature foosball table (for modelling)	1	\$36.15	2	\$72.30	y
Arducam 100fps USB camera	1	\$83.29	1	\$83.29	y
3D printed custom prints and filament	2	\$101.45	–	\$101.45	y
Cables, connectors, and wires	1	–	–	\$299.85	y
L-brackets, screws, and wood	2	–	–	\$53.71	y
Power supplies (LRS series)	1	\$50	4	\$200	y
Raspberry Pi 5 (16GB) + cooling + PSU	1	\$226.08	1	\$226.08	y
SD card (128GB)	2	\$19.20	1	\$19.20	y
Servos (NEMA 23 Easy Servo)	1	\$174.26	2	\$348.52	y
Linear belt drive (HPVB45)	1	\$152.58	4	\$610.32	y
Teknic ClearCore Controller	1	\$179.60	1	\$179.60	y
Shipping, duties, and taxes	2	–	–	\$89.72	y
Base foosball table	1	\$80.00	1	\$80.00	y
IKEA Table	2	\$60.00	1	\$60.00	y
Foosball Rod	2	\$19.99	2	\$39.98	y
Total Consumables / Services				\$2,478.70	

Table 7. Capital equipment budget table.

Capital Equipment	Priority	Cost / Unit	Quantity	Total Cost (\$)	Requires Funding (y/n)
No capital equipment (all components under consumables)	–	–	–	\$0	n
Total Capital Equipment				\$0	

Table 8. Student labour budget table.

Student Labour	Cost / Hr¹	Hours²	Total (\$)
Joonhan Ryu	\$42	120	\$5,040
Luke Watson	\$42	120	\$5,040
Daniel Yu	\$42	120	\$5,040
Weiqing Zhang	\$42	120	\$5,040
Total Student Labour (unfunded)			\$20,160

Total Cost of Project	\$22,638.70
Total Cost Requiring Funding	\$2,478.70

Funding Summary	
Students (\$100 each)	\$400
Supervisor	\$226.08
Other (sponsorships, materials reuse)	\$0
Requested from ECE496	\$1,852.62
Total Funding	\$2,478.70

6.3.2 Contingency Plan

In the event that funding is promised but delayed, the team will provide the necessary costs with the expectation of later reimbursement. In the event that funding is reduced, there are multiple scenarios depending on the amount the team is able to receive:

- **Amount received is enough to cover everything except the components required for actuation of the second foosball rod.**
 - In this case, the team will cover the costs for the components of the second foosball rod.
- **Amount received is not enough to cover the components required for the actuation of the first foosball rod.**
 - In this case, the team will fund the remaining components and limit the scope to one foosball rod.
- **Amount received is nothing.**
 - In this case, the team will fund all the components for the actuation of one foosball rod and again limit the scope.

6.4 Appendix D: Rotational Motor Specification Calculations

This appendix section shows the calculations behind the system requirement specifications for the rotational motor's speed and torque.

¹Based on cost/hr from the provided example budget table.

²Based on the Gantt chart in Figure 2 and assuming five hours of work per team member for approximately six months (Sept., Oct., Nov., Jan., Feb., Mar.).

6.4.1 Rotational Speed Calculations

Assume the tangential speed of the foosman (attached to the rod) directly translates to the speed of the ball.

Most component specifications for speed are given in rotations per minute (RPM). To derive this from speed, this formula can be used:

$$\text{RPM} = \frac{\omega \cdot 60}{2\pi}$$

where ω is the angular velocity given by $\omega = \frac{v}{r}$.

The radius in this case is the distance from the center of the rod to the foot of the foosmen, since that is the point of contact with the ball.

The height of the center of the rod from the playing field is about 3", or 76mm. On a professional table, the foosmen have a clearance of about 6.35mm. In practice this value can be much higher, and since a higher clearance will give a smaller radius, and therefore larger RPM, take a higher bound of 16mm. Therefore we can assume the radius to be approximately 60mm. Now we can graph this equation to see what RPM will be required for desired speeds as seen in Figure 3.

$$y = \frac{v \cdot 60}{2\pi \cdot 0.06}$$

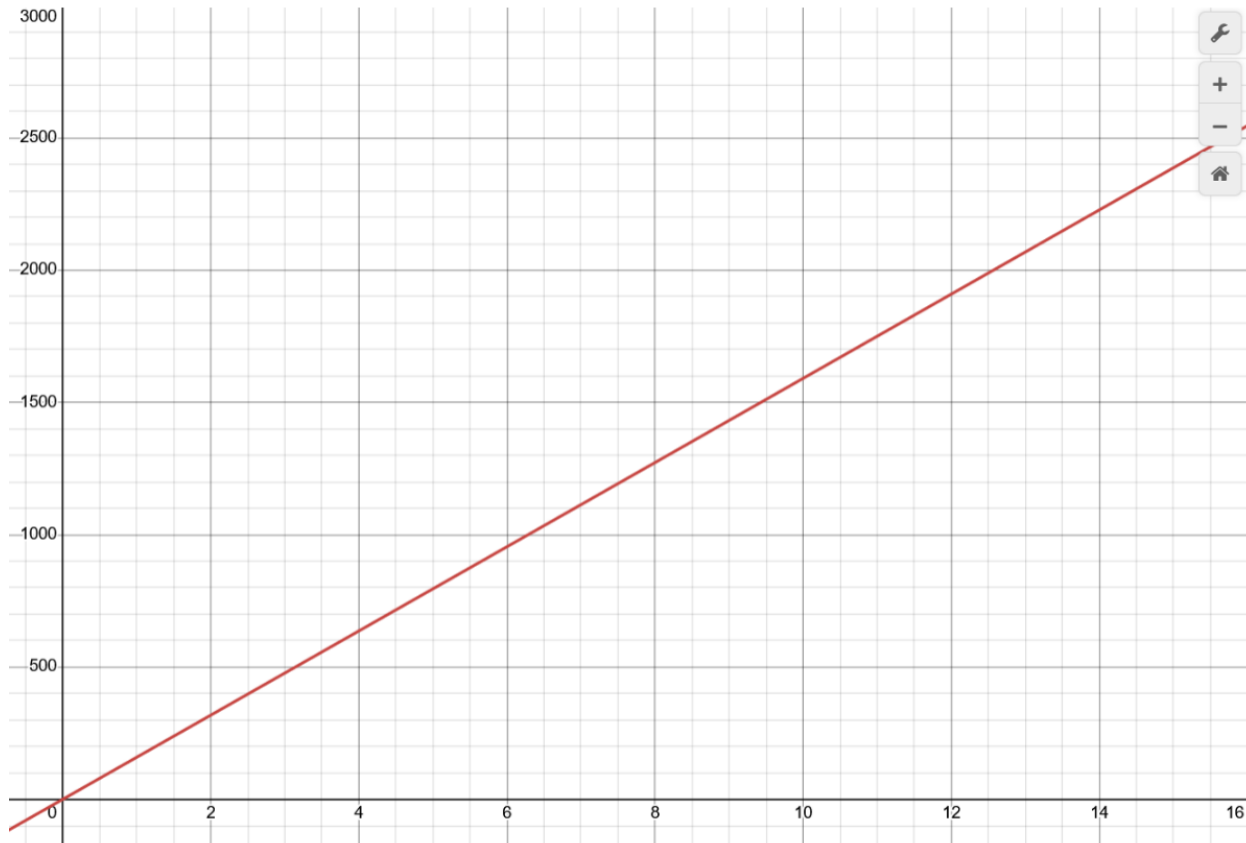


Figure 3. Speed to RPM.

Thus to achieve a speed of 6m/s, around 1000RPM is required.

6.4.2 Rotational Torque Calculations

The torque required to accelerate the rod to a desired velocity will be calculated using the formula:

$$\tau = I \cdot \alpha$$

where I is the total inertia of the system and α is the angular acceleration.

I can be calculated as the sum of the load inertia and the motor's rated inertia:

$$I = I_{load} + I_{rotor}$$

where load inertia is the combination of inertia from the rod and from the foosmen:

$$I_{load} = I_{rod} + I_{foosmen} = \frac{1}{2}m \cdot r^2 + \sum_{i=0}^N m_i \cdot r_i^2$$

Assume the mass of the rod to be 0.7kg^3 and the radius to be 8 mm^4 . This gives

$$I_{rod} = \frac{1}{2}(0.7)(0.008)^2 = 2.24 \cdot 10^{-5}\text{kg} \cdot \text{m}^2$$

Since we are focusing on two defensive lines of play, the most foosmen a rod can have is three, so we will use that value for N. Take a weight of 50g^5 and radius of 60mm , we get

$$I_{foosmen} = \sum_{i=0}^N m_i r_i^2 = N \cdot m \cdot r^2 = 3 \cdot 0.05 \cdot 0.06^2 = 5.4 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$$

and subsequently

$$I_{load} = 2.24 \cdot 10^{-5} + 5.4 \cdot 10^{-4} = 5.625 \cdot 10^{-4} \text{kg} \cdot \text{m}^2$$

The reference datasheet in Appendix G shows the rotor inertia to be $524\text{ g} \cdot \text{cm}^2$, giving

$$I = 5.265 \cdot 10^{-4} + 524 \cdot 10^{-7} = 5.79 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$$

Next, to calculate angular acceleration, we first take a target speed of 6m/s and foosmen contact radius of 0.06m to calculate angular velocity:

$$\omega = \frac{v}{r} = \frac{6}{0.06} = 100 \text{ rad/s}$$

Time is more of a limiting factor than angle due to the fact that a servo should be able to rotate the rod a full 360 degrees, so we will use the

$$\alpha = \frac{\omega}{t} \text{ instead of } \alpha = \frac{\omega^2}{2\theta}.$$

Finally, with a 50ms timeframe for the motor to reach this desired speed from rest ($\omega_0 = 0$), we get

$$\alpha = \frac{\omega}{t} = \frac{100}{0.05} = 2000 \text{ rad/s}^2$$

³Assuming a steel hollow rod.

⁴Using a typical rod with diameter of $5/8''$.

⁵Typical weight for a non-counterbalanced foosmen.

Final torque is then calculated as:

$$\tau = 5.79 \cdot 10^{-4} \cdot 2000 = 1.158 \text{ N} \cdot \text{m}$$

6.5 Appendix E: Translational Motor Specification Calculations

The goal for the translational motor is to achieve realistic gameplay in a casual setting against a human opponent. We are aiming for a 1m/s translational speed with a reaction time of 0.05s⁶. In addition, since the translational motor needs to carry rotational motor, we can assume that it needs to carry at least 2kg of weight⁷.

In order to determine the minimum required torque and speed, we first need to determine the total amount of force applied on this system using both the acceleration and force equations.

$$F_g = mg = 2 \text{ kg} \times 9.81 \frac{\text{m}}{\text{s}^2} \approx 20 \text{ N}$$

$$a = \frac{v}{t} = \frac{1 \frac{\text{m}}{\text{s}}}{0.05 \text{ s}} = 20 \frac{\text{m}}{\text{s}^2}$$

$$F_a = ma = 2 \text{ kg} \times 20 \frac{\text{m}}{\text{s}^2} = 40 \text{ N}$$

$$F_t = F_g + F_a = 20 \text{ N} + 40 \text{ N} = 60 \text{ N}$$

From our calculations above, we need to exert at least 40 Newtons of force. To determine the corresponding peak torque and speed requirements, we can apply the torque, angular velocity, and RPM formulas below.

$$\tau = rF \sin \theta = 0.02 \text{ m} \times 60 \text{ N} = 1.2 \text{ N} \cdot \text{m}$$

The sine theta angle above is 90 degrees since the linear guide rail creates a constant perpendicular angle, resulting in the sine component of the torque equation to be equal to 1.

$$\omega = \frac{v}{r} = \frac{10 \frac{\text{m}}{\text{s}}}{0.02 \text{ m}} = 50 \frac{\text{rad}}{\text{s}}$$

The radius component in both the torque and angular velocity equations is estimated to be around 20mm from research completed on current off-the-shelf linear guide rail models.

⁶Based on the same desired timeframe from standstill to full speed in Appendix D.

⁷From Appendix G, the selected motor is 1.54kg in weight and so we can assume 2kg as a safe margin of error with regards to additional miscellaneous weights such as screws.

$$\text{RPM} = \frac{\omega \times 60}{2\pi} = \frac{50 \times 60}{2\pi} \approx 500 \text{ RPM}$$

Based on our calculations, the required peak torque of our motor is 1.2Nm while the speed of the motors to maintain our goal of 1m/s needs to be at least capable of 500RPM. If needed, a gearbox can be added to this system to further increase the output torque.

6.6 Appendix F: Computer-Aided Design (CAD) Models

Figure 4 shows the computer-aided design (CAD) model for the wooden camera stand and black camera holder attached directly over the foosball table.



Figure 4. CAD model of the camera setup over the foosball table.

Figure 5 shows the CAD model of the MEAN WELL LRS-350-36 power supply unit in silver and its custom safety enclosure in black.



Figure 5. CAD model of the power supply with a custom enclosure.

Figure 6 shows the overall motor setup with the HPVB45 linear guide rail for one foosball rod. The translation motor is on the right coupled to the linear guide rail for translational movement while the rotational motor is sitting top of the carriage system. The green components are all customized 3D prints to secure the motors and the guide rail.

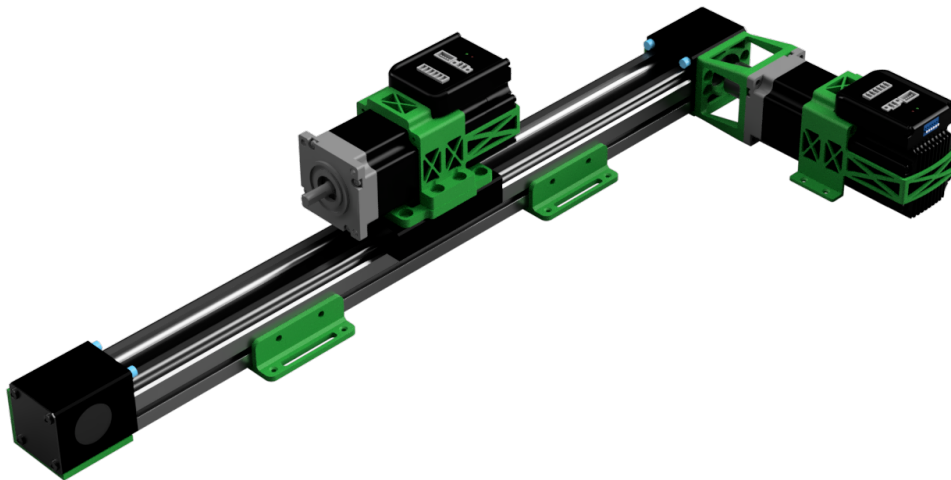


Figure 6. CAD model of our dual motor setup.

6.7 Appendix G: Servo Datasheet Reference

Figure 7 shows the specifications for the selected motor.

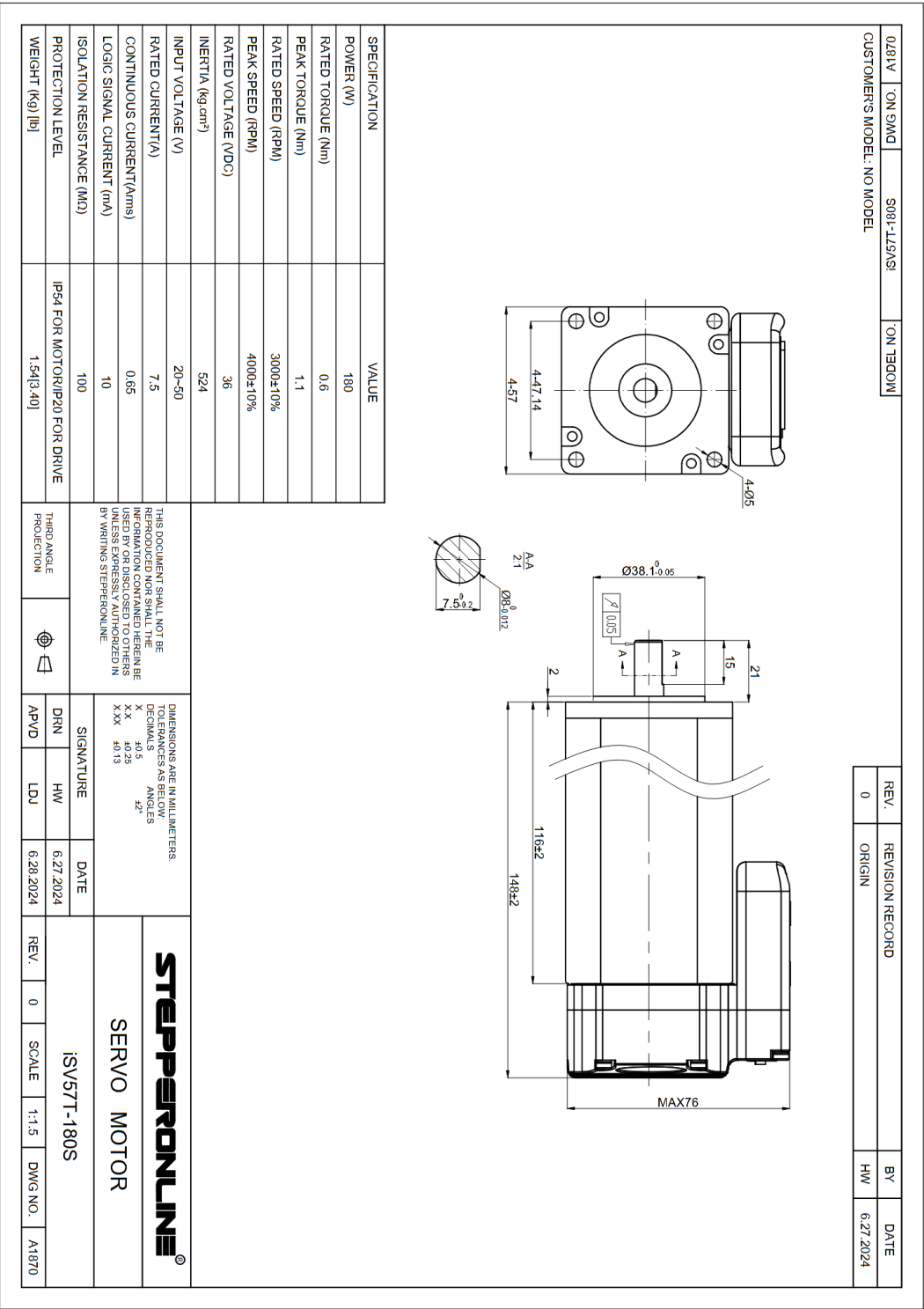


Figure 7. Selected motor datasheet [3].